The Wormhole Marginal Throat Selected by a Minimal Complex-Conjugate Scale and Its Opening Conditions

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Abstract

We investigate the opening conditions of the wormhole marginal throat selected by a minimal complex-conjugate scale. Starting from the dimensionless scale ratio $x = r/R$, the pair $\zeta_\pm = x \pm i$ generates a real geometric response through phase cancellation. This response has its maximum at the characteristic scale $r = R$. In a Morris–Thorne-type static and spherically symmetric geometry, the pure geometric sector gives $b'(R) = 1$, meaning that the response selects a marginal throat rather than an opened one. To open this throat, we introduce a throat-opening tensor constructed from the variational residual window of the scalar-field quasi-solution. The resulting local conditions lower the throat derivative below unity while keeping the redshift function finite at the throat. For the representative choice $\alpha = \delta_{\max}$, we obtain $b'(R) \simeq 0.615$, satisfying the flare-out condition.

We further show that the radial matching coefficient can be fixed by the supply peak of the normalized exchange current, giving $\beta \simeq 1.739$. With this value, the exterior matching condition admits a regular inner branch at $x_a \simeq 2.416$. The outer branch at $x_a \simeq 4.001$ is rejected because the radial metric factor vanishes earlier at $x_h \simeq 3.217$, whereas the accepted inner branch satisfies the thin-shell-free matching condition $\Phi'_{\rm in}(x_a) = \tilde{b}(x_a)/(2x_a^2 D(x_a)) = \Phi'_{\rm out}(x_a) \simeq 2.126$ in dimensionless notation. On this branch, the redshift function and curvature invariants remain finite, with the matching surface located at $a \simeq 2.416R$ ($R = GM/c^2$), between the exterior Schwarzschild radius $r_{s,\rm out} \simeq 2.201R$ and the exterior photon-sphere radius $r_{\rm ph,out} \simeq 3.302R$, satisfying $r_{s,\rm out} < a < r_{\rm ph,out}$. Thus, the locally opened throat can be matched to a Schwarzschild exterior branch while retaining the exterior photon-sphere structure. Within the static and spherically symmetric setting, this regular branch can be extended across the throat in a proper-distance coordinate, giving a symmetric two-sided opened-throat candidate with identical matching radii $a \simeq 2.416R$ and exterior Schwarzschild radii $r_{s,\rm out} \simeq 2.201R$ on both sides. These results show that the variational residual can function as an internal geometric window for opening the marginal throat selected by the minimal complex-conjugate scale, and that the resulting opened-throat region admits a regular two-sided exterior matching candidate.

I. Introduction

Wormhole geometries have long served as useful theoretical laboratories for exploring finite-radius throat structures and possible departures from classical black-hole interiors [1–4]. In classical general relativity, gravitational collapse of a sufficiently compact object leads to the formation of a black hole. In the classical description, the interior evolution approaches a curvature singularity at $r = 0$, signaling the breakdown of the classical theory. It is therefore natural to ask whether this central singularity can be replaced by a finite-radius geometric structure. Related attempts include gravastars [5, 6], regular black holes [7–10], black-bounce geometries [11–13], and quantum-corrected black-hole interiors [14–16].

A wormhole throat also provides one possible finite-radius structure. In the Morris–Thorne construction, a static and spherically symmetric geometry is described by a shape function $b(r)$ and a redshift function $\Phi(r)$ [17–19]. If the throat is located at $r = r_0$, the basic throat condition is $b(r_0) = r_0$, while the flare-out condition is $b'(r_0) < 1$, and the redshift factor must also remain finite at the throat, $0 < e^{2\Phi(r_0)} < \infty$ [17–21]. These conditions distinguish a physically opened throat from a formal radial minimum by requiring both flare-out and a finite redshift function. More general discussions of wormhole throat geometry, flare-out conditions, and energy-condition constraints can be found in Refs. [20–24].

The purpose of the present paper is not to assume an opened wormhole throat from the beginning. Instead, we ask what type of throat structure is selected by a minimal complex-conjugate scale defined in a radial one-dimensional scale sector. Starting from the dimensionless scale ratio $x = r/R$, we introduce the pair $\zeta_\pm = x \pm i$. This pair is minimal in the sense that it is the smallest complex-conjugate structure needed to generate a real geometric response through phase cancellation, without introducing an additional dimensionless imaginary-scale parameter. In this construction, the radial one-dimensional scale sector $x = r/R$ is treated as the primary structure from which the geometric response is generated, while the static and spherically symmetric spacetime geometry is used as the local geometric arena in which the resulting throat conditions are evaluated.

The key observation is that the pure geometric response derived from this minimal complex-conjugate scale does not directly produce an opened throat. For the Morris–Thorne-type geometry, the temporal component of the pure sector gives $b'(r) = r^2\Lambda(r)$. Since the response reaches $\Lambda(R) = 1/R^2$ at the characteristic scale $r = R$, one obtains $b'(R) = 1$. Thus, the pure response saturates the flare-out condition and selects a marginal throat rather than an opened one.

The central question of this paper is therefore how this marginal throat can be opened without introducing an external exotic matter source, unlike the usual exotic-matter support associated with traversable wormholes [17–19, 24]. To address this question, we introduce a throat-opening tensor $\mathcal{Q}^{\mu}{}_{\nu}$. This tensor is interpreted as an internal geometric correction constructed using the variational residual window of the scalar-field quasi-solution. This viewpoint is distinct from thin-shell constructions, where the supporting stress-energy is localized on a junction hypersurface [25–27]. It is also distinct from approaches in which quantum stress-energy, vacuum polarization, semiclassical backreaction, or Casimir-type effects are used as effective sources for wormhole support [28–38]. We derive the local opening conditions $\mathcal{Q}^{t}{}_{t}(R) < 0$ and $\mathcal{Q}^{r}{}_{r}(R) = 0$, and show that the variational residual can lower the throat derivative from $b'(R) = 1$ to $b'(R) \simeq 0.615$, satisfying the flare-out condition locally.

The paper is organized as follows. We first derive the geometric response from the minimal complex-conjugate scale and formulate the variational residual window. We then show that the pure geometric sector selects a marginal throat and derive the local conditions required to open it. Finally, we construct the throat-opening tensor from the residual window, analyze the exterior matching branch, and discuss the resulting static two-sided opened-throat candidate.

This organization also reflects the scope of the present work. We do not claim a fully dynamical traversable wormhole solution in this paper. Rather, we show that the minimal complex-conjugate scale selects a wormhole marginal throat, and that a variational-residual-based throat-opening tensor provides the local conditions under which this marginal throat can be opened. Within the static and spherically symmetric setting, the resulting opened-throat branch can be extended to a symmetric two-sided candidate and matched to a regular Schwarzschild exterior branch. Schwarzschild-like wormholes provide useful comparison cases for this type of exterior matching [1, 3, 4], while black-hole mimicker and dark-compact-object phenomenology provides the broader observational context [39]. A full dynamical stability analysis under time-dependent perturbations is left for future work, together with possible extensions to wave scattering and echo phenomenology [40–44], as well as gray-body-factor calculations and related numerical tools [45].

II. Minimal complex-conjugate scale and geometric response

We begin with the radial one-dimensional scale ratio

$$x = \frac{r}{R},$$

where R denotes the positive characteristic length scale of each physical regime, while both r and R are treated as dimensionless quantities until calibrated against observations. The dimensionless minimal complex-conjugate scale pair is defined by

$$\zeta_{\pm} = x \pm i.$$

This pair is symmetric with respect to the real x-axis and introduces the smallest imaginary displacement away from the real scale sector.

The real background response is obtained by adding the logarithms of the two conjugate branches,

$$\mathcal{B}(x) = \ln \zeta_{+} + \ln \zeta_{-}.$$

Since $\zeta_{+}\zeta_{-} = 1 + x^2$, this gives

$$\mathcal{B}(x) = \ln(1 + x^2).$$

Thus, the imaginary phases cancel, and the remaining background response is real.

The scalar profile is defined as the radial derivative of this real background response,

$$\phi(r) = \frac{d}{dr} \ln\left(1 + \frac{r^2}{R^2}\right).$$

Therefore,

$$\phi(r) = \frac{2r}{R^2 + r^2}.$$

This profile is regular at $r = 0$, preserves the characteristic scale R, and decreases as $2/r$ for $r \gg R$.

The radius-dependent geometric response is defined by

$$\Lambda(r) = \phi(r)^2.$$

Thus,

$$\Lambda(r) = \frac{4r^2}{(R^2 + r^2)^2}.$$

Equivalently, using $x = r/R$, this can be written as

$$\Lambda(r) = \frac{1}{R^2} F(x), \qquad F(x) = \frac{4x^2}{(1 + x^2)^2}.$$

The function $F(x)$ reaches its maximum at $x = 1$, where

$$F(1) = 1.$$

Therefore,

$$\Lambda(R) = \frac{1}{R^2}.$$

This peak relation $\Lambda(R) = 1/R^2$ is the key input for the throat analysis below.

In the black-hole interpretation, the same characteristic scale can be identified with the critical radius at which the inward gravitational acceleration is balanced by the scale-dependent repulsive response. Let the repulsive acceleration associated with the geometric response be proportional to the scalar profile,

$$a_{\rm rep}(r) = c_0\phi(r) = \frac{2c_0 r}{R^2 + r^2}.$$

Here, c_0 is a characteristic scale coefficient that converts the scalar profile into an acceleration, with dimension $\mathrm{m}^2/\mathrm{s}^2$, equivalent to an energy per unit mass. This acceleration reaches its maximum at $r = R$. We therefore identify the critical radius with the characteristic scale,

$$r_{\rm crit} = R.$$

At the critical radius, the repulsive acceleration is

$$a_{\rm rep}(R) = \frac{c_0}{R}.$$

Balancing this with the Newtonian gravitational acceleration GM/R^2 gives

$$\frac{c_0}{R} = \frac{GM}{R^2},$$

and therefore

$$c_0 = \frac{GM}{R}.$$

If the repulsive energy scale at the collapse-stopping radius is identified with the black-hole mass-energy scale, one has $c_0 = c^2$. Hence,

$$R = \frac{GM}{c^2}.$$

Thus, in the black-hole interior interpretation,

$$r_{\rm crit} = R = \frac{GM}{c^2} = \frac{1}{2}r_s, \qquad r_s = \frac{2GM}{c^2},$$

where r_s is the Schwarzschild radius [46].

This identification does not mean that $R = GM/c^2$ is already an opened throat. It only fixes the characteristic black-hole interior scale selected by the geometric response. As shown later, the pure geometric sector first gives a marginal throat at this scale, and the opening condition must be supplied separately.

III. Variational residual window

In this section, the scalar profile $\phi(r) = 2r/(R^2+r^2)$ in the one-dimensional radial scale sector is substituted into the corresponding standard one-dimensional Lagrangian. The resulting deviation from the Euler–Lagrange extremum condition is read as the variational residual encoding the internal response tension.

We use the standard kinetic-minus-potential form

$$\mathcal{L}_\phi = -\frac{1}{2}\left(\frac{d\phi}{dr}\right)^2 - V(\phi).$$

The potential is chosen as the minimal even polynomial containing the leading quadratic response and the first nonlinear correction,

$$V(\phi) = -\phi^2 + A\phi^4.$$

The coefficient A is fixed by requiring the quadratic and quartic contributions to belong to the same characteristic $4/R^2$ hierarchy after substituting the scale-preserving profile.

Introducing $x = r/R$, the quadratic contribution becomes

$$\phi(r)^2 = \frac{4r^2}{(R^2+r^2)^2} = \frac{4}{R^2}\frac{x^2}{(1+x^2)^2}.$$

Thus, the leading quadratic response belongs to the overall $4/R^2$ hierarchy.

The quartic contribution is

$$\phi(r)^4 = \left[\frac{4}{R^2}\frac{x^2}{(1+x^2)^2}\right]^2 = \left(\frac{4}{R^2}\right)^2\left[\frac{x^2}{(1+x^2)^2}\right]^2.$$

This term contains one additional factor of $4/R^2$ compared with the quadratic response. Therefore, to place the quartic correction in the same $4/R^2$ hierarchy as the leading term, we choose

$$A = \frac{1}{4}R^2.$$

With this choice,

$$A\phi(r)^4 = \frac{1}{4}R^2\left(\frac{4}{R^2}\right)^2\left[\frac{x^2}{(1+x^2)^2}\right]^2 = \frac{4}{R^2}\left[\frac{x^2}{(1+x^2)^2}\right]^2.$$

Hence, the quartic correction is placed in the same overall $4/R^2$ hierarchy as the leading quadratic term.

The scalar potential is therefore fixed as

$$V(\phi) = -\phi^2 + \frac{1}{4}R^2\phi^4.$$

After substituting the scalar profile, this becomes

$$V(\phi(r)) = -\frac{4}{R^2}\frac{x^2}{(1+x^2)^2} + \frac{4}{R^2}\left[\frac{x^2}{(1+x^2)^2}\right]^2.$$

The first term is the leading response $-\Lambda(r)$, while the second term is the first nonlinear correction within the same $4/R^2$ scale hierarchy.

The corresponding Euler–Lagrange extremum condition (with r and R still being dimensionless quantities) is

$$\frac{d^2\phi}{dr^2} = R^2\phi^3 - 2\phi.$$

We define the variational residual by

$$\delta(r) \equiv \frac{d^2\phi}{dr^2} - \left(R^2\phi^3 - 2\phi\right).$$

Substituting the scalar profile into this definition gives

$$\delta(r) = \frac{4r\left(R^4 - 3R^2 + r^4 + r^2\right)}{\left(R^2 + r^2\right)^3}.$$

This residual measures the deviation of the scale-preserving scalar profile from the Euler–Lagrange extremum condition. In this sense, it can also be viewed as a reduced, dimensionless composition profile encoding the tension between the kinetic-type and potential-type response components. It is used to localize the internal opening correction.

To select a representative dimensionless residual profile, we require the residual to have a smoother onset at the origin by eliminating its linear term. The first derivative at the origin is

$$\left.\frac{d\delta}{dr}\right|_{r=0} = \frac{4(R^2 - 3)}{R^4}.$$

Thus, imposing

$$\left.\frac{d\delta}{dr}\right|_{r=0} = 0$$

gives

$$R^2 = 3.$$

Taking the positive branch gives $R = \sqrt{3}$, and we denote this selected reduced-coordinate value by $R_{(\sqrt{3})} \equiv \sqrt{3}$. With this value, the residual becomes

$$\delta(r) = \frac{4r^3(r^2 + 1)}{(r^2 + 3)^3}.$$

This form isolates a finite, localized residual structure while keeping the characteristic scale explicit.

Numerically, this residual has the values

$$\delta(R_{(\sqrt{3})}) \simeq 0.385, \qquad \delta_{\max} \simeq 0.641.$$

The maximum occurs at approximately $r \simeq 3.56$. Therefore, the residual is finite at the throat scale and reaches its peak outside the throat.

Using the relation $x = r/R$, the normalized residual window is defined by

$$W_\delta(x) = \left.\frac{\delta(Rx)}{\delta_{\max}}\right|_{R=R_{(\sqrt{3})}}.$$

At the throat scale $x = 1$, this gives

$$W_\delta(1) = \frac{\delta(R_{(\sqrt{3})})}{\delta_{\max}} \simeq \frac{0.385}{0.641} \simeq 0.601.$$

The role of $W_\delta(x)$ is to specify where the opening correction is localized. Since $W_\delta(x)$ is dimensionless, the product $W_\delta(x)\Lambda(r)$ has the same curvature dimension as the Einstein tensor. Thus, the variational residual window can be used to construct a throat-opening tensor without treating the residual itself as an independently introduced exotic matter source. The residual window constructed here will be used below as the localization structure of the throat-opening tensor.

IV. Marginal throat selected by the pure geometric sector

We now examine the throat structure selected by the pure geometric response. In this section, no throat-opening tensor is introduced. The purpose is to determine what kind of throat is obtained from $\Lambda(r)$ alone.

We use the Morris–Thorne-type static and spherically symmetric line element [17, 19]

$$ds^2 = -e^{2\Phi(r)}c^2dt^2 + \frac{dr^2}{1 - b(r)/r} + r^2 d\Omega^2.$$

Here, $b(r)$ is the shape function and $\Phi(r)$ is the redshift function. If the throat is located at $r = r_0$, the throat condition is

$$b(r_0) = r_0.$$

In the present construction, the characteristic scale selected by the geometric response is identified with the throat candidate,

$$r_0 = R.$$

Thus, the throat condition becomes

$$b(R) = R.$$

The pure geometric sector is written as

$$G^\mu{}_\nu + \Lambda(r)\delta^\mu{}_\nu = 0.$$

For the Morris–Thorne metric, the temporal mixed component is

$$G^t{}_t = -\frac{b'(r)}{r^2}.$$

Therefore, the temporal component of the pure sector gives

$$-\frac{b'(r)}{r^2} + \Lambda(r) = 0.$$

Equivalently,

$$b'(r) = r^2\Lambda(r).$$

Evaluating this relation at the candidate throat $r = R$, we obtain

$$b'(R) = R^2\Lambda(R).$$

Since the response generated by the minimal complex-conjugate scale satisfies $\Lambda(R) = 1/R^2$, this gives

$$b'(R) = 1.$$

The Morris–Thorne flare-out condition for an opened throat [17, 20, 21] is

$$b'(R) < 1.$$

The pure geometric response therefore does not satisfy the strict flare-out inequality. Instead, it reaches the boundary value $b'(R) = 1$, saturating the flare-out condition as an equality.

Thus, the pure geometric response selects a marginal throat at $r = R$. It determines the throat scale but does not open the throat by itself.

V. Action notation for the opening tensor

Before specifying the local throat-opening components, we formulate the opening tensor in an action notation adapted to the static and spherically symmetric geometry. In this notation, the opening tensor is generated by a scalar quantity whose metric variation produces the anisotropic temporal, radial, and transverse responses required below.

We first express the same dimensionless scale variable x geometrically by using the area $\mathcal{A}$ of the symmetry two-spheres,

$$x = \frac{1}{R}\sqrt{\frac{\mathcal{A}}{4\pi}}.$$

In the areal-radius coordinate used in the following sections, $\mathcal{A} = 4\pi r^2$, and therefore this definition reduces to $x = r/R$.

Since the radial sector enters $\mathcal{Q}_{\mu\nu}$ only through $s_\mu s_\nu$, we use the local representation of the radial unit covector

$$s_\mu = \frac{\nabla_\mu x}{\sqrt{g^{\lambda\eta}\nabla_\lambda x \nabla_\eta x}}.$$

With this representation, the normalization condition is

$$g^{\mu\nu} s_\mu s_\nu = 1.$$

In the static sector, the temporal unit vector is defined from the static Killing vector ξ^μ as

$$u^\mu = \frac{\xi^\mu}{\sqrt{-\xi^\lambda \xi_\lambda}}.$$

It satisfies

$$u^\mu u_\mu = -1.$$

The static and spherically symmetric background also gives the orthogonality condition

$$u^\mu s_\mu = 0.$$

The vectors u^μ and s^μ define the local temporal and radial directions of the static spherical geometry. The mixed components of $\mathcal{Q}^\mu{}_\nu$ are treated as scalar functions of x in this sector. A diagonal anisotropic opening tensor compatible with spherical symmetry can therefore be written in the projector form

$$\mathcal{Q}_{\mu\nu} = \mathcal{Q}^\theta{}_\theta g_{\mu\nu} - \left(\mathcal{Q}^t{}_t - \mathcal{Q}^\theta{}_\theta\right) u_\mu u_\nu - \left(\mathcal{Q}^\theta{}_\theta - \mathcal{Q}^r{}_r\right) s_\mu s_\nu.$$

In the mixed-index form adapted to the static spherical background, this expression gives

$$\mathcal{Q}^\mu{}_\nu = \mathrm{diag}\left(\mathcal{Q}^t{}_t, \mathcal{Q}^r{}_r, \mathcal{Q}^\theta{}_\theta, \mathcal{Q}^\theta{}_\theta\right).$$

Thus the temporal, radial, and transverse opening responses are organized by the natural projectors of the static spherical sector.

We now introduce an action notation that generates this projector form. Let

$$S_Q = \frac{1}{\kappa}\int d^4x\sqrt{-g}\,\Xi,$$

where Ξ is the scalar generating quantity for the opening tensor in the static spherical sector. We take

$$\Xi = \mathcal{Q}^\theta{}_\theta + \frac{1}{2}\left(\mathcal{Q}^t{}_t - \mathcal{Q}^\theta{}_\theta\right)(I_u + 1) + \frac{1}{2}\left(\mathcal{Q}^\theta{}_\theta - \mathcal{Q}^r{}_r\right)(I_s - 1),$$

with

$$I_u = g^{\mu\nu} u_\mu u_\nu, \qquad I_s = g^{\mu\nu} s_\mu s_\nu.$$

On the static spherical background, these invariants reduce to

$$I_u = -1, \qquad I_s = 1.$$

Therefore the background value of the generating scalar is

$$\Xi_0 = \mathcal{Q}^\theta{}_\theta.$$

For this static-sector action notation, the algebraic metric variation is taken with respect to the projector invariants I_u and I_s. The opening tensor is defined by the metric variation of S_Q,

$$\mathcal{Q}_{\mu\nu} = -\frac{2\kappa}{\sqrt{-g}}\frac{\delta S_Q}{\delta g^{\mu\nu}}.$$

Since Ξ depends algebraically on the metric through I_u and I_s in the present sector, this becomes

$$\mathcal{Q}_{\mu\nu} = g_{\mu\nu}\Xi - 2\frac{\partial \Xi}{\partial g^{\mu\nu}}.$$

Using

$$\frac{\partial I_u}{\partial g^{\mu\nu}} = u_\mu u_\nu,$$

and

$$\frac{\partial I_s}{\partial g^{\mu\nu}} = s_\mu s_\nu,$$

one obtains, on the static spherical background,

$$\mathcal{Q}_{\mu\nu} = \mathcal{Q}^\theta{}_\theta g_{\mu\nu} - \left(\mathcal{Q}^t{}_t - \mathcal{Q}^\theta{}_\theta\right) u_\mu u_\nu - \left(\mathcal{Q}^\theta{}_\theta - \mathcal{Q}^r{}_r\right) s_\mu s_\nu.$$

Thus the anisotropic opening tensor used below is represented by a scalar action notation in the static and spherically symmetric sector.

At this stage, the three scalar components $\mathcal{Q}^t{}_t(x)$, $\mathcal{Q}^r{}_r(x)$, and $\mathcal{Q}^\theta{}_\theta(x)$ have not yet been specified. The next section derives the local throat conditions on the temporal and radial components. The residual-window construction then determines the minimal forms of $\mathcal{Q}^t{}_t(x)$ and $\mathcal{Q}^r{}_r(x)$, while the transverse component $\mathcal{Q}^\theta{}_\theta(x)$ is fixed by the covariant conservation of the combined effective geometric sector.

A full time-dependent perturbation theory would require specifying how the temporal and radial projectors are extended away from the static sector. That issue is left for future work together with the dynamical stability analysis.

VI. Local opening conditions at the marginal throat

Having formulated the opening tensor in an action notation adapted to the static and spherically symmetric sector, we now derive the local conditions required for this tensor to open the marginal throat selected by the pure geometric response.

As shown above, the pure sector gives $b'(R) = 1$ at the marginal throat located at $r = R$. Hence the Morris–Thorne flare-out condition [17, 20, 21] is saturated rather than strictly satisfied. The role of the opening tensor $\mathcal{Q}^\mu{}_\nu$ is to deform this marginal configuration into a locally opened throat.

We write the corrected geometric field equation as

$$G^\mu{}_\nu + \Lambda(r)\delta^\mu{}_\nu + \mathcal{Q}^\mu{}_\nu(r) = 0,$$

leaving the detailed functional form of the throat-opening tensor $\mathcal{Q}^\mu{}_\nu$ unspecified at this stage. We only derive the local conditions that this tensor must satisfy at the throat.

For the Morris–Thorne-type metric, the temporal mixed component is

$$G^t{}_t = -\frac{b'(r)}{r^2}.$$

Therefore, the temporal component of the corrected equation gives

$$-\frac{b'(r)}{r^2} + \Lambda(r) + \mathcal{Q}^t{}_t(r) = 0.$$

Equivalently,

$$b'(r) = r^2 \left[\Lambda(r) + \mathcal{Q}^t{}_t(r)\right].$$

At the candidate throat $r = R$, using $\Lambda(R) = 1/R^2$, this becomes

$$b'(R) = 1 + R^2\mathcal{Q}^t{}_t(R).$$

To open the marginal throat, the flare-out condition requires $b'(R) < 1$. Hence,

$$1 + R^2\mathcal{Q}^t{}_t(R) < 1,$$

and therefore

$$\mathcal{Q}^t{}_t(R) < 0.$$

This is the first local opening condition. It states that the temporal component of the opening tensor must give a negative correction at the throat in order to lower $b'(R)$ below unity.

Next, we consider the redshift function. The radial component of the corrected equation gives

$$-\frac{b(r)}{r^3} + 2\left(1 - \frac{b(r)}{r}\right)\frac{\Phi'(r)}{r} + \Lambda(r) + \mathcal{Q}^r{}_r(r) = 0.$$

Solving this relation for $\Phi'(r)$, we obtain

$$\Phi'(r) = \frac{r}{2\left(1 - b(r)/r\right)}\left[\frac{b(r)}{r^3} - \Lambda(r) - \mathcal{Q}^r{}_r(r)\right].$$

At the throat, $b(R) = R$, so the denominator $1 - b(R)/R$ vanishes. For the redshift function to remain finite at the throat, the numerator must vanish at the same point [17, 19]. Evaluating the numerator at $r = R$, we find

$$\frac{b(R)}{R^3} - \Lambda(R) - \mathcal{Q}^r{}_r(R) = \frac{1}{R^2} - \frac{1}{R^2} - \mathcal{Q}^r{}_r(R).$$

Thus, the regularity condition reduces to

$$\mathcal{Q}^r{}_r(R) = 0.$$

This is the second local opening condition. It ensures that the redshift function does not acquire a divergence at the throat.

The two local conditions required to open the marginal throat are therefore

$$\mathcal{Q}^t{}_t(R) < 0, \qquad \mathcal{Q}^r{}_r(R) = 0.$$

The first condition controls the flare-out inequality, while the second controls the finiteness of the redshift function. The following section constructs a concrete form of $\mathcal{Q}^\mu{}_\nu$ from the variational residual window and shows that these two conditions can be satisfied simultaneously.

VII. Opening by the variational residual window

We now construct the throat-opening tensor using the variational residual window introduced above. The purpose is to deform the marginal throat selected by the pure geometric sector into a locally opened throat.

The required local opening conditions are $\mathcal{Q}^t{}_t(R) < 0$ and $\mathcal{Q}^r{}_r(R) = 0$. A minimal form satisfying the first condition is obtained by choosing the temporal component as

$$\mathcal{Q}^t{}_t(r) = -\alpha W_\delta(x)\Lambda(r),$$

where α is a dimensionless coefficient controlling the strength of the temporal opening component, with $0 \le \alpha \le \delta_{\max}$. At the throat $x = 1$, this gives

$$\mathcal{Q}^t{}_t(R) = -\alpha W_\delta(1)\Lambda(R) = -\frac{\alpha W_\delta(1)}{R^2}.$$

Therefore, for $\alpha > 0$ and $W_\delta(1) > 0$, the condition $\mathcal{Q}^t{}_t(R) < 0$ is satisfied.

The corrected temporal equation gives

$$b'(r) = r^2\left[\Lambda(r) + \mathcal{Q}^t{}_t(r)\right].$$

Evaluating this relation at $r = R$, and using $\Lambda(R) = 1/R^2$, one obtains

$$b'(R) = 1 - \alpha W_\delta(1).$$

Thus, the marginal value $b'(R) = 1$ is lowered below unity whenever $\alpha W_\delta(1) > 0$. If one additionally requires $0 < b'(R) < 1$, then the coefficient must satisfy $0 < \alpha W_\delta(1) < 1$.

For the radial component, the redshift regularity condition requires $\mathcal{Q}^r{}_r(R) = 0$. We use the form

$$\mathcal{Q}^r{}_r(r) = -\beta\left(1 - \frac{1}{x}\right) W_\delta(x)\Lambda(r),$$

where β is a dimensionless coefficient controlling the strength of the radial opening component. Since $1 - 1/x = 0$ at $x = 1$, this component automatically satisfies

$$\mathcal{Q}^r{}_r(R) = 0.$$

Therefore, the temporal component opens the throat, while the radial component is constructed so as to preserve the finiteness of the redshift function at the throat.

Equivalently, using $\Lambda(r) = R^{-2}F(x)$, these components can be written in dimensionless form as

$$R^2\mathcal{Q}^t{}_t(x) = -\alpha W_\delta(x)F(x),$$

and

$$R^2\mathcal{Q}^r{}_r(x) = -\beta\left(1 - \frac{1}{x}\right) W_\delta(x)F(x).$$

We now evaluate the opening strength. Since $W_\delta(1) = \delta(R_{(\sqrt{3})})/\delta_{\max}$, a natural choice is to identify the opening coefficient with the maximum residual strength,

$$\alpha = \delta_{\max}.$$

Then

$$\alpha W_\delta(1) = \delta_{\max}\frac{\delta(R_{(\sqrt{3})})}{\delta_{\max}} = \delta(R_{(\sqrt{3})}),$$

and therefore

$$b'(R) = 1 - \delta(R_{(\sqrt{3})}).$$

Using $\delta(R_{(\sqrt{3})}) \simeq 0.385$, we obtain

$$b'(R) \simeq 0.615.$$

Hence,

$$0 < b'(R) < 1$$

is satisfied.

This result shows that the variational residual window can deform the marginal throat into a locally opened throat. This residual window is not interpreted as an independently introduced exotic matter source. Rather, it acts as a dimensionless localization window multiplying the geometric response $\Lambda(r)$. In this sense, the quasi-solution character of the scalar profile supplies the internal geometric correction required to open the marginal throat.

The functional form of the transverse component $\mathcal{Q}^\theta{}_\theta$ is also not specified independently, but is fixed by covariant conservation, as discussed later.

VIII. Exchange current and radial matching coefficient

The radial component of the throat-opening tensor was introduced with a dimensionless coefficient β. In this section, we fix this coefficient from the exchange-current structure of the scale-dependent geometric response, rather than treating it as an independent free parameter.

The geometric response can be written in the universal form

$$\Lambda(r) = \frac{1}{R^2}F(x), \qquad F(x) = \frac{4x^2}{(1+x^2)^2}, \qquad x = \frac{r}{R}.$$

Since $\Lambda(r)$ depends on the radial coordinate, its radial gradient defines an exchange current associated with the scale-dependent geometric sector. In the static and spherically symmetric case, we write

$$J_r = \frac{1}{\kappa}\frac{d\Lambda}{dr},$$

where $\kappa = 8\pi G/c^4$. Using $x = r/R$, one obtains

$$\frac{d\Lambda}{dr} = \frac{1}{R^3}\frac{dF}{dx},$$

and therefore

$$J_r = \frac{1}{\kappa R^3}\frac{dF}{dx}.$$

The dimensionless exchange current is then obtained by normalizing J_r,

$$\tilde{J}(x) \equiv \kappa R^3 J_r = \frac{dF}{dx},$$

and explicitly

$$\tilde{J}(x) = \frac{8x(1-x^2)}{(1+x^2)^3}.$$

This function changes sign at $x = 1$. For $x < 1$, $\tilde{J}(x) > 0$, while for $x > 1$, $\tilde{J}(x) < 0$. Thus, $x = 1$ is the transition point between the supply and recovery branches of the exchange-current structure.

The extrema of the normalized exchange current are determined by

$$\frac{d\tilde{J}}{dx} = \frac{d^2F}{dx^2} = 0.$$

The second derivative is

$$\frac{d^2F}{dx^2} = \frac{8(3x^4 - 8x^2 + 1)}{(1+x^2)^4}.$$

Therefore, the extrema satisfy

$$3x^4 - 8x^2 + 1 = 0.$$

Solving this equation gives

$$x_{\rm sup} = \sqrt{\frac{4-\sqrt{13}}{3}}, \qquad x_{\rm rec} = \sqrt{\frac{4+\sqrt{13}}{3}}.$$

Numerically,

$$x_{\rm sup} \simeq 0.363, \qquad x_{\rm rec} \simeq 1.592.$$

Substitution into $\tilde{J}(x)$ yields

$$\tilde{J}(x_{\rm sup}) \simeq +1.739, \qquad \tilde{J}(x_{\rm rec}) \simeq -0.443.$$

The positive extremum at $x_{\rm sup}$ represents the maximum supply branch of the exchange current. We use this supply peak to fix the radial matching coefficient,

$$\beta = \tilde{J}_{\rm sup}^{\rm max} \simeq 1.739.$$

With this prescription, β is not introduced as an arbitrary parameter. It is determined by the maximum exchange capacity of the scale-dependent response itself.

The recovery peak $\tilde{J}(x_{\rm rec}) \simeq -0.443$ characterizes the outer recovery branch, where the exchange current changes sign and the effective correction is returned to the geometric sector. In the present throat-opening construction, the supply peak fixes the radial matching strength, while the recovery branch provides the scale at which the correction begins to relax toward the exterior region.

Finally, we note that the exchange current is balanced over the full radial domain. Since

$$\Lambda(0) = \Lambda(\infty) = 0,$$

one has

$$\int_0^\infty J_r\, dr = \frac{1}{\kappa}\left[\Lambda(\infty) - \Lambda(0)\right] = 0.$$

Thus, the one-dimensional radial exchange is balanced between the supply and recovery branches.

In addition to this radial balance, the corresponding spherical flux is

$$\Phi_J(r) = \oint_{S_r} J^i\, d\Sigma_i = 4\pi r^2 J_r.$$

Using

$$J_r = \frac{1}{\kappa}\frac{d}{dr}\left[\frac{4r^2}{(R^2+r^2)^2}\right] = \frac{1}{\kappa}\frac{8r(R^2-r^2)}{(R^2+r^2)^3},$$

we obtain

$$\Phi_J(r) = \frac{32\pi}{\kappa}\frac{r^3(R^2-r^2)}{(R^2+r^2)^3}.$$

The endpoint limits are

$$\Phi_J(0) = 0, \qquad \Phi_J(\infty) = 0.$$

Therefore, since $\Phi_J(0) = \Phi_J(\infty) = 0$, the spherical flux has no net endpoint contribution. In this sense, the exchange current represents an internal redistribution between the scale-dependent geometric response and the opening correction through the supply and recovery branches.

At sufficiently large characteristic scales, the exchange current is strongly suppressed. For a cosmological scale $R \sim 10^{26}\,\mathrm{m}$, the estimate $|d\Lambda/dr| \sim R^{-3} \sim 10^{-78}\,\mathrm{m}^{-3}$ shows that $\Lambda(r)$ can be treated as approximately constant. Hence, in the large-scale limit, the local exchange current becomes negligible and the usual Friedmann dynamics with a constant Λ is effectively recovered [47, 48].

IX. Covariant conservation and transverse response

In the present construction, the throat-opening tensor is interpreted as an effective correction to the geometric response, rather than as an independently assumed matter source. Since the scale-dependent response $\Lambda(r)$ depends on the radial coordinate, the term $\Lambda(r)\delta^\mu{}_\nu$ is not separately conserved in general. Therefore, we impose covariant conservation on the combined effective geometric sector,

$$\nabla_\mu \left[\Lambda(r)\delta^\mu{}_\nu + \mathfrak{Q}^\mu{}_\nu\right] = 0.$$

Equivalently, the opening tensor satisfies

$$\nabla_\mu \mathfrak{Q}^\mu{}_\nu = -\partial_\nu \Lambda(r).$$

Thus, $\mathfrak{Q}^\mu{}_\nu$ can be interpreted as an effective geometric correction that compensates for the radial variation of $\Lambda(r)$, rather than as an independently conserved matter tensor.

For a static and spherically symmetric correction sector, we take the opening tensor to be diagonal,

$$\mathfrak{Q}^\mu{}_\nu = \mathrm{diag}\left(\mathfrak{Q}^t{}_t,\, \mathfrak{Q}^r{}_r,\, \mathfrak{Q}^\theta{}_\theta,\, \mathfrak{Q}^\theta{}_\theta\right).$$

The two angular components are equal by spherical symmetry, so there is only one independent transverse component.

The nontrivial conservation equation is the radial component. It gives

$$\frac{d}{dr}\left(\Lambda + \mathfrak{Q}^r{}_r\right) + \Phi'(r)\left(\mathfrak{Q}^r{}_r - \mathfrak{Q}^t{}_t\right) + \frac{2}{r}\left(\mathfrak{Q}^r{}_r - \mathfrak{Q}^\theta{}_\theta\right) = 0.$$

Solving this relation for the transverse component yields

$$\mathfrak{Q}^\theta{}_\theta = \mathfrak{Q}^r{}_r + \frac{r}{2}\left[\frac{d}{dr}\left(\Lambda + \mathfrak{Q}^r{}_r\right) + \Phi'(r)\left(\mathfrak{Q}^r{}_r - \mathfrak{Q}^t{}_t\right)\right].$$

This expression shows that the angular component is not freely prescribed once $\mathfrak{Q}^t{}_t$, $\mathfrak{Q}^r{}_r$, and $\Phi(r)$ are specified. It is fixed by the covariant conservation of the combined effective sector.

At the throat, the radial opening component satisfies $\mathfrak{Q}^r{}_r(R) = 0$. Also, since $\Lambda(r)$ reaches its maximum at $r = R$, one has $d\Lambda/dr|_R = 0$. Using

$$\mathfrak{Q}^t{}_t(R) = -\frac{\alpha W_\delta(1)}{R^2}, \qquad \left.\frac{d\mathfrak{Q}^r{}_r}{dr}\right|_R = -\frac{\beta W_\delta(1)}{R^3},$$

the transverse component at the throat becomes

$$\mathfrak{Q}^\theta{}_\theta(R) = \frac{W_\delta(1)}{2R^2}\left[-\beta + \alpha R\Phi'(R)\right].$$

Equivalently, since

$$R\Phi'(R) = \left.\frac{d\Phi}{dx}\right|_{x=1},$$

this throat value can be written in the dimensionless form

$$R^2\mathfrak{Q}^\theta{}_\theta(R) = \frac{W_\delta(1)}{2}\left[-\beta + \alpha \left.\frac{d\Phi}{dx}\right|_{x=1}\right].$$

This formula gives a local diagnostic of the transverse opening response. Since $W_\delta(1)$, α, and β are finite, a finite throat limit of $d\Phi/dx$ implies that $\mathfrak{Q}^\theta{}_\theta(R)$ is also finite. Hence, the transverse component is not independently prescribed, but is fixed by covariant conservation once the temporal and radial opening components are specified, and this conservation condition does not require a singular or divergent transverse response at the opened throat. It is important to note that the transverse component need not vanish or have a positive sign. The only requirement is the finite dimensionless condition

$$\left|R^2\mathfrak{Q}^\theta{}_\theta(R)\right| < \infty.$$

X. Exterior matching and regularity

This section analyzes the matching of the locally opened throat region to an exterior Schwarzschild geometry. Using the black-hole interior identification $R = GM/c^2$ obtained above, let the matching surface be located at

$$a = Rx_a,$$

and define the dimensionless shape function by

$$\tilde{b}(x) = \frac{b(r)}{R}.$$

The throat condition is $\tilde{b}(1) = 1$. In the interior region, the temporal component of the corrected geometric equation gives

$$\tilde{b}'(x) = x^2F(x)\left[1 - \alpha W_\delta(x)\right],$$

and therefore

$$\tilde{b}(x) = 1 + \int_1^x u^2F(u)\left[1 - \alpha W_\delta(u)\right]du.$$

We take the exterior region to be described by the Schwarzschild line element [46],

$$ds^2_{\rm out} = -\left(1 - \frac{2GM_{\rm out}}{c^2r}\right)c^2dt^2 + \frac{dr^2}{1 - \frac{2GM_{\rm out}}{c^2r}} + r^2d\Omega^2.$$

Continuity of the radial metric component at $r = a$ requires

$$b_{\rm in}(a) = \frac{2GM_{\rm out}}{c^2}.$$

Thus the exterior mass is determined by

$$M_{\rm out} = \frac{c^2}{2G}b_{\rm in}(a) = \frac{c^2R}{2G}\tilde{b}(x_a).$$

If the internal reference mass is defined by $R = GM/c^2$, then

$$\frac{M_{\rm out}}{M} = \frac{\tilde{b}(x_a)}{2}.$$

The time component of the first fundamental form gives

$$e^{2\Phi_{\rm in}(a)} = 1 - \frac{2GM_{\rm out}}{c^2a} = 1 - \frac{\tilde{b}(x_a)}{x_a}.$$

Therefore, the matching surface must satisfy

$$D(x_a) \equiv 1 - \frac{\tilde{b}(x_a)}{x_a} > 0.$$

In contrast to thin-shell junction constructions, where the surface stress-energy is determined by junction conditions [25–27], the non-Schwarzschild radial effective contribution must vanish at the matching surface:

$$\Lambda(a) + \mathfrak{Q}^r{}_r(a) = 0.$$

Using

$$\mathfrak{Q}^r{}_r(r) = -\beta\left(1 - \frac{1}{x}\right)W_\delta(x)\Lambda(r),$$

this becomes

$$\Lambda(a)\left[1 - \beta\left(1 - \frac{1}{x_a}\right)W_\delta(x_a)\right] = 0.$$

Since $\Lambda(a) \neq 0$ at finite a, this condition reduces to

$$\beta\left(1 - \frac{1}{x_a}\right)W_\delta(x_a) = 1.$$

With $\beta = \tilde{J}^{\rm max}_{\rm sup} \simeq 1.739$, this equation admits two numerical solutions,

$$x_a \simeq 2.416, \qquad x_a \simeq 4.001.$$

However, not both branches are physically suitable. The radial metric factor is controlled by

$$D(x) = 1 - \frac{\tilde{b}(x)}{x}.$$

Numerically, $D(x)$ vanishes at

$$x_h \simeq 3.217.$$

Therefore, the outer branch $x_a \simeq 4.001$ lies beyond this zero and is not suitable as a regular matching surface for the opened-throat interior.

The inner branch satisfies

$$x_a \simeq 2.416 < x_h \simeq 3.217.$$

Hence $D(x) > 0$ is maintained throughout $1 < x \leq x_a$. At the matching surface on this branch, one finds

$$D(x_a) \simeq 0.0887 > 0.$$

Thus, the interior opened-throat region can be matched to a Schwarzschild exterior at

$$a \simeq 2.416R$$

without crossing an internal zero of the radial metric factor.

The regularity of this inner branch can be further checked by evaluating the redshift function and the main curvature invariants. From the radial field equation, the redshift derivative is

$$\Phi'(r) = \frac{r}{2[1 - b(r)/r]} \left[\frac{b(r)}{r^3} - \Lambda(r) - \mathcal{Q}^r{}_r(r) \right].$$

Equivalently, in terms of $x = r/R$, this equation becomes

$$\frac{d\Phi}{dx} = \frac{x}{2D(x)} \left[\frac{\tilde{b}(x)}{x^3} - F(x) - R^2 \mathcal{Q}^r{}_r(x) \right],$$

with $D(x) = 1 - \tilde{b}(x)/x$. At the throat, both the numerator and denominator vanish consistently because $\tilde{b}(1) = 1$, $F(1) = 1$, and $R^2\mathcal{Q}^r{}_r(1) = 0$. In particular, near the throat, both vanish linearly in $x-1$, while the denominator has leading coefficient $2D'(1)$, with $D'(1) > 0$. Therefore, the redshift derivative has a finite throat limit. Numerically, for the regular inner branch, one finds

$$\left. \frac{d\Phi}{dx} \right|_{x=1^+} \simeq -1.741.$$

The redshift function also remains finite from the throat to the matching surface. The integrated change is

$$\Phi(x_a) - \Phi(1) \simeq 0.635.$$

The additive constant of Φ is fixed by the time-component matching condition,

$$e^{2\Phi(x_a)} = D(x_a),$$

or equivalently,

$$\Phi(x_a) = \frac{1}{2} \ln D(x_a).$$

The interior profile is then obtained by integrating inward,

$$\Phi(x) = \Phi(x_a) - \int_x^{x_a} \frac{d\Phi}{du}\, du.$$

In terms of the dimensionless coordinate x, the interior metric is equivalently written as

$$ds^2 = -e^{2\Phi(x)} c^2 dt^2 + \frac{R^2 dx^2}{D(x)} + R^2 x^2 d\Omega^2.$$

Since this metric is smooth for $1 < x < x_a$, it is then sufficient to evaluate the curvature invariants at the endpoint limits. Denoting the Ricci scalar by $\mathcal{R}$, one obtains

$$\mathcal{R}(1^+) \simeq \frac{1.90}{R^2}, \qquad \mathcal{R}(x_a) \simeq -\frac{1.00}{R^2}.$$

The Ricci tensor squared remains finite, with

$$R_{\mu\nu}R^{\mu\nu}(1^+) \simeq \frac{1.42}{R^4}, \qquad R_{\mu\nu}R^{\mu\nu}(x_a) \simeq \frac{0.743}{R^4}.$$

The Kretschmann scalar also remains finite:

$$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}(1^+) \simeq \frac{4.75}{R^4}, \qquad R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}(x_a) \simeq \frac{1.28}{R^4}.$$

Thus, the inner matching branch does not develop a curvature singularity in the interval $1 \leq x \leq x_a$. Together with $D(x) > 0$ for $1 < x \leq x_a$, these finite curvature invariants support the interpretation of this branch as a locally regular opened-throat region matched to a Schwarzschild exterior. The matching surface lies outside the exterior Schwarzschild radius determined through the internal shape function, while the detailed behavior of the photon sphere and the associated shadow scale can be examined separately.

XI. Two-sided extension and global matching candidate

We now examine whether the accepted inner branch can be extended across the throat to form a two-sided static geometry matched to Schwarzschild exterior regions on both sides.

For the representative inner branch, the numerical values relevant to the regular two-sided extension are

$$\tilde{b}'(1) \simeq 0.615 < 1, \qquad D'(1) \simeq 0.385 > 0,$$

and, at the matching surface,

$$\tilde{b}(x_a) \simeq 2.201, \qquad D(x_a) \simeq 0.0887 > 0.$$

Thus, the accepted branch satisfies $D(1) = 0$ at the throat and $D(x) > 0$ for $1 < x \leq x_a$. Hence, $D(x)$

has no additional horizon-like zero between the throat and the matching surface.

As discussed above, the non-Schwarzschild radial effective contribution must vanish at the matching surface:

$$\Lambda(a) + \mathcal{Q}^r{}_r(a) = 0.$$

Using $\Lambda(a) = R^{-2}F(x_a)$, this condition can be written in dimensionless form as

$$F(x_a) + R^2\mathcal{Q}^r{}_r(a) = 0.$$

On the other hand, the interior radial equation gives

$$\Phi'_{\rm in}(x_a) = \frac{x_a}{2D(x_a)}\left[\frac{\tilde{b}(x_a)}{x_a^3} - F(x_a) - R^2\mathcal{Q}^r{}_r(a)\right].$$

Substituting the matching-surface condition into the interior expression, the last two terms cancel, and one obtains

$$\Phi'_{\rm in}(x_a) = \frac{\tilde{b}(x_a)}{2x_a^2 D(x_a)}.$$

For the exterior Schwarzschild region, the first fundamental form gives

$$r_{s,\rm out} = R\tilde{b}(x_a),$$

and therefore

$$\Phi'_{\rm out}(x_a) = \frac{\tilde{b}(x_a)}{2x_a^2 D(x_a)}.$$

Thus,

$$\Phi'_{\rm in}(x_a) = \frac{\tilde{b}(x_a)}{2x_a^2 D(x_a)} = \Phi'_{\rm out}(x_a).$$

For the accepted matching radius $x_a \simeq 2.416$, this yields

$$\Phi'_{\rm in}(x_a) = \Phi'_{\rm out}(x_a) \simeq 2.126.$$

Therefore, the accepted inner branch satisfies the thin-shell-free matching condition, with the interior and exterior redshift gradients agreeing at the matching surface with the finite value $\simeq 2.126$.

A convenient coordinate for describing the two-sided extension is the proper radial distance l, defined by

$$\frac{dl}{dr} = \pm\left(1 - \frac{b(r)}{r}\right)^{-1/2}.$$

Introducing the dimensionless variables

$$x = \frac{r}{R}, \qquad \tilde{b}(x) = \frac{b(r)}{R}, \qquad D(x) = 1 - \frac{\tilde{b}(x)}{x},$$

the proper-distance relation becomes

$$\frac{dl}{R} = \pm\frac{dx}{\sqrt{D(x)}}.$$

The two signs correspond to the two sides of the throat. We choose the origin of l at the throat, so that

$$x = 1, \qquad l = 0.$$

At this point,

$$D(1) = 0.$$

The flare-out condition obtained above gives

$$\tilde{b}'(1) < 1.$$

Since $D(x) = 1 - \tilde{b}(x)/x$, one finds at the throat

$$D'(1) = 1 - \tilde{b}'(1) > 0.$$

Therefore, near $x = 1$,

$$D(x) \simeq D'(1)(x-1).$$

Substituting this into the proper-distance relation gives

$$l \simeq \pm 2R\sqrt{\frac{x-1}{D'(1)}}.$$

Equivalently,

$$x - 1 \simeq \frac{D'(1)}{4}\left(\frac{l}{R}\right)^2.$$

Thus, the areal radius

$$r(l) = Rx(l)$$

is an even function of l near the throat and has a regular minimum at

$$l = 0.$$

In particular,

$$\left.\frac{dr}{dl}\right|_{l=0} = 0, \qquad \left.\frac{d^2r}{dl^2}\right|_{l=0} > 0.$$

This gives the local proper-distance representation of the opened throat.

Since the inner branch satisfies

$$D(x) > 0 \qquad (1 < x \le x_a),$$

the proper distance from the throat to the matching surface is finite:

$$l_a = R\int_1^{x_a}\frac{dx}{\sqrt{D(x)}}.$$

Numerically, for the inner matching branch, this gives a finite value of order

$$l_a \simeq 5.5R.$$

Therefore, in the proper-distance coordinate, the interior region is represented as

$$-l_a \leq l \leq l_a,$$

with the throat at $l = 0$.

Following the standard picture of a symmetric two-sided wormhole throat [17, 19], a simple symmetric extension is obtained by taking

$$r(l) = r(-l), \qquad \Phi(l) = \Phi(-l).$$

The interior metric can then be written in the proper-distance form

$$ds^2 = -e^{2\Phi(l)}c^2dt^2 + dl^2 + r(l)^2 d\Omega^2.$$

In this form, the two sides correspond to

$$l > 0$$

and

$$l < 0.$$

Each side reaches a matching surface at

$$l = \pm l_a.$$

At these two surfaces, the interior geometry may be matched to exterior Schwarzschild regions with the same exterior mass parameter fixed by

$$M_{\rm out} = \frac{c^2 R}{2G}\tilde{b}(x_a).$$

Taken together, these results indicate that the throat is not supported by an additional thin shell or singular matter layer placed at the throat itself. In the proper-distance coordinate, the throat appears simply as a regular point where the areal radius reaches its minimum. Furthermore, since the thin-shell-free condition is satisfied at the matching surfaces $r = a = Rx_a$ on both sides, the two interior branches may be smoothly connected to Schwarzschild exterior regions.

Thus, within the static and spherically symmetric setting considered here, the opened-throat region admits a natural two-sided extension in proper-distance coordinates. This provides a consistent static two-sided traversable-throat candidate, rather than a fully established dynamical wormhole solution. In particular, the time-dependent stability of this two-sided configuration remains to be analyzed separately.

XII. Discussion

The analysis developed above shows that the minimal complex-conjugate scale selects a marginal throat through the pure geometric response. In the pure sector, the temporal equation gives $b'(R) = 1$, and therefore the Morris–Thorne flare-out condition is saturated. Thus the characteristic scale $r = R$ is not an arbitrary throat radius, but the threshold point at which the pure response reaches the marginal throat state.

In the black-hole interpretation, the same characteristic scale can be identified with $R = GM/c^2$. At the level of the pure response, this scale corresponds to a marginal, non-opened throat. The condition is $b'(R) = 1$, not $b'(R) < 1$. Therefore, $R = GM/c^2$ should be understood as the location of a threshold throat selected by the pure geometric response. It is not, by itself, a fully opened throat.

The throat-opening tensor $\mathfrak{Q}^{\mu}{}_{\nu}$ supplies the internal correction required to move the geometry away from this marginal state. Its temporal component lowers the throat derivative below unity, while its radial component is required to vanish at the throat in order to keep the redshift function finite. These two requirements are expressed by

$$\mathfrak{Q}^{t}{}_{t}(R) < 0, \qquad \mathfrak{Q}^{r}{}_{r}(R) = 0.$$

The first condition controls the flare-out inequality, and the second controls the local regularity of the redshift function.

In the present construction, the throat-opening tensor is built from the variational residual window. The residual $\delta(r)$ measures the deviation of the scalar-field profile from the exact Euler–Lagrange extremum condition. After normalization, it defines the window function $W_\delta(x)$, which localizes the opening correction. With the representative choice $\alpha = \delta_{\rm max}$, the throat derivative becomes

$$b'(R) = 1 - \delta(R_{(\sqrt{3})}) \simeq 0.615.$$

This satisfies

$$0 < b'(R) < 1,$$

so the marginal throat is locally opened at the level of the Morris–Thorne flare-out condition.

The radial sector provides an additional consistency structure. The radial matching coefficient β is fixed by the supply peak of the normalized exchange current, giving

$$\beta = \tilde{J}^{\rm max}_{\rm sup} \simeq 1.739.$$

With this value, the matching condition admits an inner solution at

$$x_a \simeq 2.416.$$

This solution is selected over the outer branch since the radial metric factor remains positive beyond the throat up to the matching surface. In particular, the zero of $D(x) = 1 - \tilde{b}(x)/x$ occurs at $x_h \simeq 3.217$, while the inner matching radius satisfies $x_a < x_h$. Thus, $1 < x \leq x_a$ has no horizon-like zero of the radial metric factor.

For the same inner branch, the redshift function remains finite up to the matching surface. The curvature invariants evaluated in the throat region are also finite. In particular, the Ricci scalar, the Ricci tensor squared, and the Kretschmann scalar remain finite both near the throat and at $x_a \simeq 2.416$. Therefore, the resulting geometry can be interpreted as a locally nonsingular opened throat region, rather than merely as a formal flare-out condition.

It is useful to clarify how this opened sector would appear under an effective right-hand-side interpretation. Although the opening sector is treated here as a geometric contribution, one may formally move it to the right-hand side and read it as an effective stress-energy contribution:

$$\kappa T^{\mu}{}_{\nu,\mathrm{eff}} = -\Lambda(r)\delta^{\mu}{}_{\nu} - \mathfrak{Q}^{\mu}{}_{\nu}.$$

With the mixed-component convention

$$T^{\mu}{}_{\nu} = \mathrm{diag}(-\rho, p_r, p_t, p_t),$$

and with the explicit r-dependence suppressed, this gives

$$\rho_{\mathrm{eff}} = \frac{\Lambda + \mathfrak{Q}^{t}{}_{t}}{\kappa}, \quad p_{r,\mathrm{eff}} = -\frac{\Lambda + \mathfrak{Q}^{r}{}_{r}}{\kappa}, \quad p_{t,\mathrm{eff}} = -\frac{\Lambda + \mathfrak{Q}^{\theta}{}_{\theta}}{\kappa}.$$

Therefore, the radial null combination is

$$\rho_{\mathrm{eff}} + p_{r,\mathrm{eff}} = \frac{\mathfrak{Q}^{t}{}_{t} - \mathfrak{Q}^{r}{}_{r}}{\kappa}.$$

At the throat, using the local opening conditions $\mathfrak{Q}^{t}{}_{t}(R) < 0$ and $\mathfrak{Q}^{r}{}_{r}(R) = 0$, one obtains

$$\rho_{\mathrm{eff}}(R) + p_{r,\mathrm{eff}}(R) < 0.$$

For this inner branch, we define the dimensionless diagnostics by

$$\bar{\rho} = \kappa R^2 \rho_{\mathrm{eff}}, \qquad \bar{p}_r = \kappa R^2 p_{r,\mathrm{eff}}, \qquad \bar{p}_t = \kappa R^2 p_{t,\mathrm{eff}}.$$

Numerically, these quantities satisfy

$$\bar{\rho} > 0$$

throughout $1 \leq x \leq x_a$. The radial null combination obeys

$$\bar{\rho} + \bar{p}_r < 0$$

only in a near-throat region, crosses zero at $x \simeq 1.584$, located just before the recovery peak $x_{\mathrm{rec}} \simeq 1.592$ ($\tilde{J}(x_{\mathrm{rec}}) \simeq -0.443$), and remains non-negative up to the matching surface at $x_a \simeq 2.416$. By contrast, the transverse null combination, computed using the conservation-fixed $\mathfrak{Q}^{\theta}{}_{\theta}$, satisfies

$$\bar{\rho} + \bar{p}_t > 0$$

throughout the same interval. These diagnostics show that, under a formal right-hand-side interpretation, the opening sector can play the effective role usually assigned to exotic matter support. However, in the present formulation, this sector is consistently retained on the geometric side of the field equations, since it is constructed from the internally generated geometric response rather than introduced as an independent matter source.

The exterior interpretation is more delicate. For the inner matching branch, the exterior Schwarzschild radius is determined by the internal shape function as

$$\frac{r_{s,\mathrm{out}}}{R} = \tilde{b}(x_a).$$

With $\tilde{b}(x_a) \simeq 2.201$, this gives

$$r_{s,\mathrm{out}} \simeq 2.201R, \qquad a \simeq 2.416R.$$

The matching surface lies outside the exterior Schwarzschild radius. At the same time, the exterior photon-sphere radius is

$$\frac{r_{\mathrm{ph,out}}}{R} = \frac{3}{2}\tilde{b}(x_a),$$

and therefore

$$r_{\mathrm{ph,out}} \simeq 3.302R.$$

Hence,

$$r_{s,\mathrm{out}} < a < r_{\mathrm{ph,out}}.$$

The photon sphere therefore remains in the exterior Schwarzschild region. This means that the throat-opening construction does not immediately remove the usual photon-sphere-controlled shadow structure.

In terms of the internal scale R, the corresponding Schwarzschild shadow diameter is

$$\frac{D_{\mathrm{sh}}}{R} = 3\sqrt{3}\,\tilde{b}(x_a).$$

This expression follows from the standard Schwarzschild photon-sphere and shadow-scale relation [49, 50]. For the accepted inner branch, this gives

$$\frac{D_{\mathrm{sh}}}{R} \simeq 11.44.$$

This increase relative to the standard Schwarzschild value $6\sqrt{3} \simeq 10.39$ [50] reflects the fact that the exterior mass inferred from the matching is larger than the internal reference mass, with $M_{\mathrm{out}} \simeq 1.10M$. This value is not used here as a direct observational fit, but it gives the characteristic shadow scale associated with the accepted matching branch. Its role is to show that the matching branch places the photon sphere outside the matching surface and remains compatible with a Schwarzschild-like exterior shadow region, a quantity of observational interest in horizon-scale imaging [51–53].

The distinction between a static two-sided throat candidate and a dynamically stable wormhole solution should be kept explicit. The analysis above supports the former, but not the latter. The remaining task is to determine whether this static two-sided configuration is dynamically stable under time-dependent perturbations, and whether its exterior matching leads to distinguishable wave-scattering or echo signatures [40–44].

XIII. CONCLUSION

In this paper, we studied the wormhole marginal throat selected by the minimal complex-conjugate scale $\zeta_\pm = x \pm i$, with $x = r/R$, and derived the local conditions required to open it. The complex-conjugate pair produces the real background response $\mathcal{B}(x) = \ln\zeta_+ + \ln\zeta_- = \ln(1+x^2)$. From this response, the scalar profile $\phi(r) = 2r/(R^2+r^2)$ is obtained, and the radius-dependent geometric response $\Lambda(r) = \phi(r)^2 = 4r^2/(R^2+r^2)^2$ follows. Thus, $\Lambda(r)$ is not introduced as an arbitrary radial function, but is generated by the minimal complex-conjugate scale.

The response $\Lambda(r)$ reaches its maximum at the characteristic scale $r = R$, where $\Lambda(R) = 1/R^2$. For the Morris–Thorne-type geometry, the pure geometric sector $G^\mu{}_\nu + \Lambda(r)\delta^\mu{}_\nu = 0$ gives $b'(r) = r^2\Lambda(r)$ from the temporal component. Evaluating this relation at $r = R$, one obtains $b'(R) = R^2\Lambda(R) = 1$. Therefore, the pure geometric response selected by the minimal complex-conjugate scale does not directly produce an opened throat. Instead, it reaches the boundary value $b'(R) = 1$, saturating the flare-out condition as an equality, and forms a marginal throat.

To open this marginal throat, we introduced the throat-opening tensor $\mathcal{Q}^\mu{}_\nu$. With this tensor, the corrected geometric equation is written as $G^\mu{}_\nu + \Lambda(r)\delta^\mu{}_\nu + \mathcal{Q}^\mu{}_\nu = 0$. The local opening conditions at the throat are $\mathcal{Q}^t{}_t(R) < 0$ and $\mathcal{Q}^r{}_r(R) = 0$. The first condition lowers $b'(R)$ below unity and realizes the flare-out condition. The second condition prevents the redshift function from diverging at the throat.

The opening tensor was constructed using the variational residual of the scalar-field quasi-solution. Defining the normalized residual window by $W_\delta(x) = (\delta(Rx)/\delta_{\max})|_{R=R_{(\sqrt{3})}}$, we used the temporal component $\mathcal{Q}^t{}_t(r) = -\alpha W_\delta(x)\Lambda(r)$, with $0 \le \alpha \le \delta_{\max}$. Then the throat derivative becomes $b'(R) = 1 - \alpha W_\delta(1)$. Choosing $\alpha = \delta_{\max}$, this reduces to $b'(R) = 1 - \delta(R_{(\sqrt{3})})$. Using the numerical value $\delta(R_{(\sqrt{3})}) \simeq 0.385$, we obtain $b'(R) \simeq 0.615$. Thus, $0 < b'(R) < 1$ is satisfied, and the marginal throat is locally opened by the opening tensor constructed from the variational residual window.

The radial matching analysis further shows that the radial coefficient is fixed by the exchange-current supply peak. Explicitly, $\tilde{J}(x) = 8x(1-x^2)/(1+x^2)^3$, $x_{\rm sup} \simeq 0.363$, and $\beta = \tilde{J}(x_{\rm sup}) \simeq 1.739$. With this value, the regular inner matching branch appears at $x_a \simeq 2.416$. Along this branch, the radial metric factor $D(x)$ remains positive up to the matching surface, and the redshift function and curvature invariants remain finite. With the black-hole interior identification $R = GM/c^2$, the matching surface lies outside the exterior Schwarzschild radius and inside the exterior photon-sphere radius,

$$r_{s,\rm out} < a < r_{\rm ph,out}, \quad 2.201R < 2.416R < 3.302R.$$

These results support the interpretation of the construction as a locally nonsingular opened-throat region with a possible Schwarzschild exterior connection.

The same inner branch also satisfies the smooth matching condition between the interior and exterior redshift derivatives,

$$\Phi'_{\rm in}(x_a) = \frac{\tilde{b}(x_a)}{2x_a^2 D(x_a)} = \Phi'_{\rm out}(x_a).$$

Thus, the accepted branch can be connected to a Schwarzschild exterior without introducing a thin shell at the matching surface, with the redshift gradients agreeing at the finite value $\simeq 2.126$.

Crucially, the accepted inner branch admits a natural two-sided extension in the proper-distance coordinate. In this description, the throat appears as a regular minimum of the areal radius, and the two sides correspond to the branches $l > 0$ and $l < 0$. In particular, at the throat,

$$\left.\frac{dr}{dl}\right|_{l=0} = 0, \qquad \left.\frac{d^2r}{dl^2}\right|_{l=0} > 0.$$

For the symmetric two-sided extension, the two exterior branches have identical matching radii, Schwarzschild radii, and exterior mass parameters, with $a_+ = a_- \simeq 2.416R$, $r_{s,+} = r_{s,-} \simeq 2.201R$, and $M_+ = M_- = M_{\rm out}$. Thus, within the present static and spherically symmetric setting, the accepted branch gives a consistent two-sided traversable-throat candidate matched to Schwarzschild exterior regions on both sides.

In this construction, the minimal complex-conjugate scale first determines the real geometric response, and the pure part of that response selects the marginal throat. The variational residual then supplies the localization structure needed to build the opening tensor. In other words, the tension encoded in the residual between the kinetic-type and potential-type response components serves as an internal geometric window for opening the marginal throat.

The present work does not claim a fully established dynamical wormhole solution. The result established here is a static and spherically symmetric construction: the minimal complex-conjugate scale selects a marginal throat, the variational-residual-based opening tensor opens it, the exchange-current supply supports the exterior matching, and the proper-distance construction allows a static two-sided extension. A full dynamical stability analysis under time-dependent perturbations remains an important subject for future work. Such an analysis would first require formulating the (t,r)-dependent opening sector,

$$\Phi(t,r), \qquad b(t,r), \qquad \mathcal{Q}^\mu{}_\nu(t,r),$$

and then perturbing it around the static background to test whether the resulting linearized system is free of growing modes, with the causal bound $R(t) \ge GM/c^2$.

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